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Bio-Inspired Robotics for Adaptive and Intelligent Autonomous Drones

Bio-inspired robotics seeks to develop intelligent machines by drawing inspiration from the principles underlying biological systems. Animals demonstrate remarkable abilities to perceive their environment, adapt their behavior, and perform complex tasks with limited computational resources. These capabilities emerge from the tight coupling between neural computation and control, morphological adaptation, and environmental interaction, which together form the foundation of embodied intelligence (see Fig. 1). Translating these principles into robotic systems has the potential to significantly enhance autonomy, adaptability, and energy efficiency, enabling robots to operate robustly in complex and dynamic environments. My research interest focuses on developing bio-inspired robotic systems that integrate neural control and adaptive mechanical design to enable robust autonomous behavior. Specifically, the research aims to investigate how biologically inspired neural control architecture and morphological intelligence can be combined to improve perception, decision-making, and sensorimotor coordination in autonomous robots.

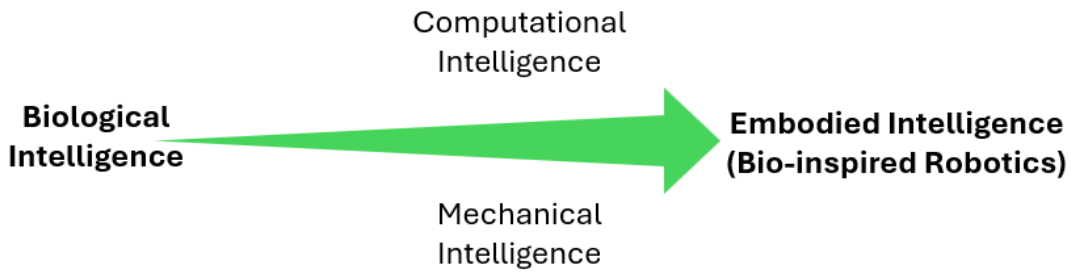

Fig. 1: *Conceptual framework of bio-inspired robotics. Biological intelligence in natural systems inspires robotic design. By integrating computational intelligence (e.g., neural control and algorithms) with mechanical intelligence (e.g., morphology, compliance, and structural adaptation), these principles are translated into embodied intelligence in robotic systems, enabling adaptive and intelligent behavior through body–environment interaction.*

During my Ph.D. study, I utilized the principles of neural dynamics, control, and learning systems for bio-inspired robotic navigation and exploration, using autonomous drones as an experimental platform.I developed a compact and modular neural control architecture capable of generating adaptive behaviors without heavy computation or extensive training, including adaptive search and exploration [1, 2], obstacle avoidance [2, 3], and speed adaptation [4] systems (see Fig. 2). I investigated these neural control systems using artificial (non-spiking) neuron models, emphasizing hysteresis, recurrence, adaptive neural dynamics, and Online learning as mechanisms for general, robust, and adaptive control on simulated and real drones (see Fig. 3A – 3C). In addition to demonstrating the generality of the proposed neural control systems, we

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deployed them on both a simulated legged robot (see Fig. 3D), and a real wheeled robot using microcontroller (ESP32) as a main computational machine (see Fig.3E). Through these works, I demonstrated that compact and modular neural control systems can be effectively deployed for real-time robotic navigation and exploration tasks, including Lévy flight–based foraging, obstacle avoidance, speed adaptation, and goal-oriented navigation. In addition to neural control systems, I also participated in research on mechanical intelligence in bio-inspired robotics. This work focused on the design of compliant and adaptive mechanical structures that enhance physical interaction

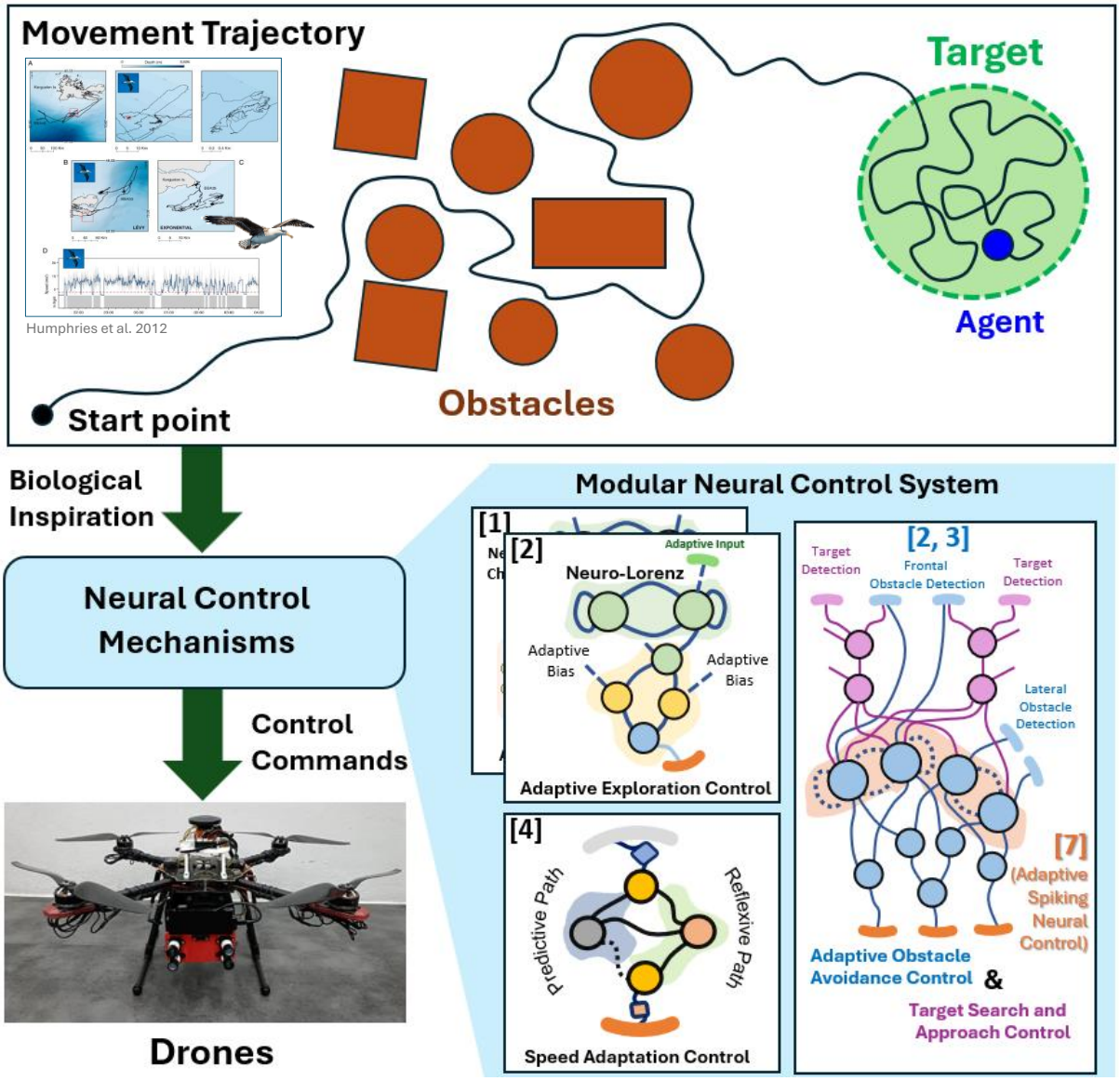

Fig. 2: *The overall block diagram outlines the process of using animal movement trajectory as a basis for developing a modular neural control system consisting of adaptive search and exploration [1, 2], obstacle avoidance [2, 3], and speed adaptation [4] modules. The system is subsequently implemented and tested utilizing drones as the experimental platform.*

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between robots and their environment. In particular, I contributed to the development of a drone landing mechanism that integrates passive compliance and structural adaptation to enable stable landing on pipe surfaces [5]. Additionally, we developed a passive–active bio-inspired compliant landing gear that enables drones to land and perch on diverse surfaces, as well as grasp objects of various shapes [6]. This study highlights how morphological intelligence can improve robustness and adaptability in robotic systems without requiring additional complex computational control systems.

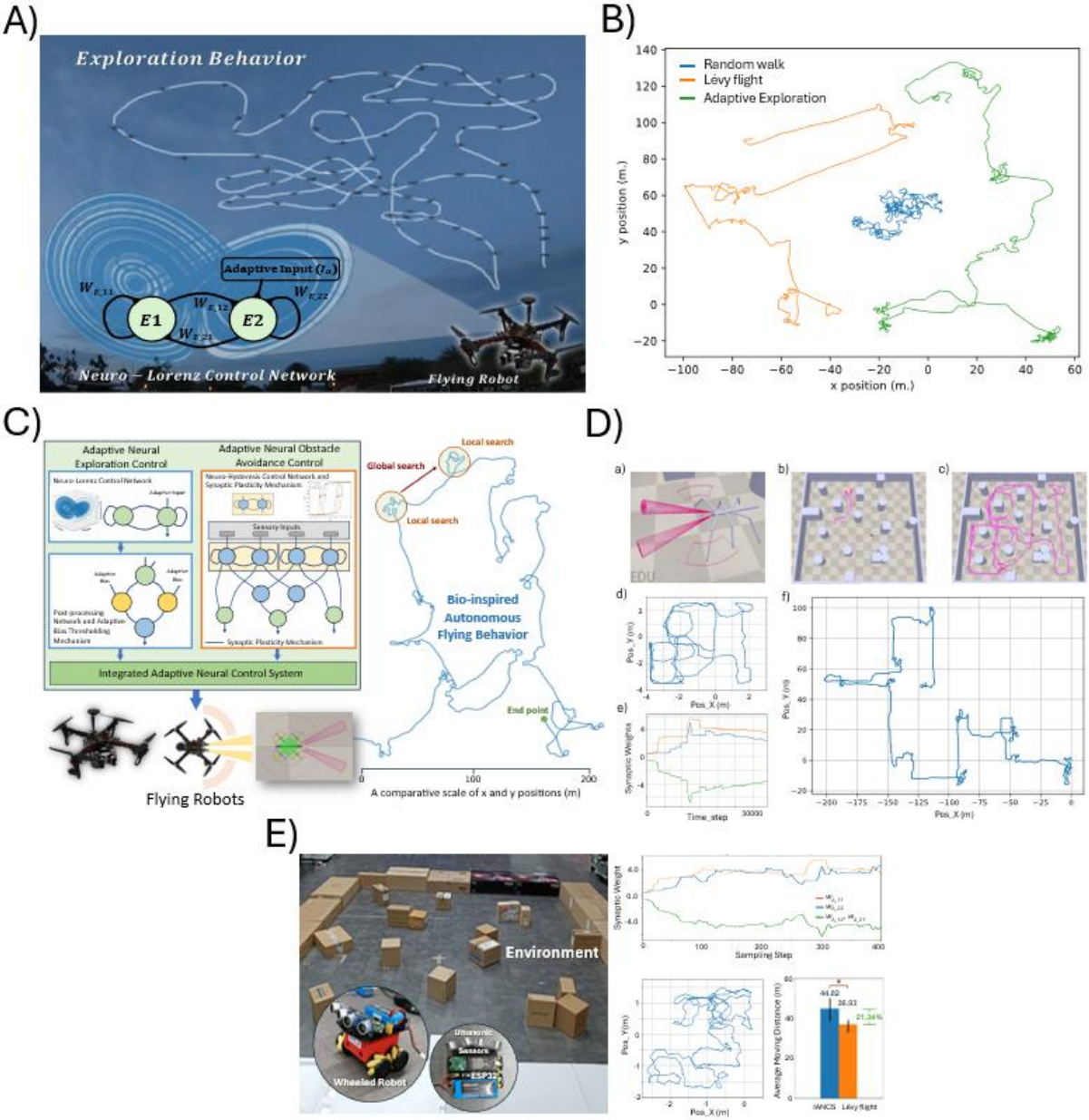

Fig. 3: *(A)-(C) Experimental results tested on simulated and real drones, (D) a simulated legged robot, and (E) a real wheeled robot using microcontroller (ESP32) as a main computational machine.*

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Furthermore, building upon my knowledge of neural dynamics, control, and learning systems in non-spiking neuron models, I extended this work to investigate spiking neuron model (Leaky Integrate-and-Fire), aiming to advance biologically plausible computation and neuromorphic approaches for real-time processing and robotic control during my visiting Ph.D. at ZHAW. In this work, I explored spiking neural dynamics and synaptic plasticity mechanisms [7] by first analyzing the dynamical properties of a recurrent spiking neuron through variations in sensory inputs and recurrent synaptic weights. I then investigated how the neural dynamics can be modulated through synaptic plasticity with online learning and additional bias inputs, and subsequently utilized to control autonomous robots (see Fig. 4). The results showed that a single recurrent spiking neuron

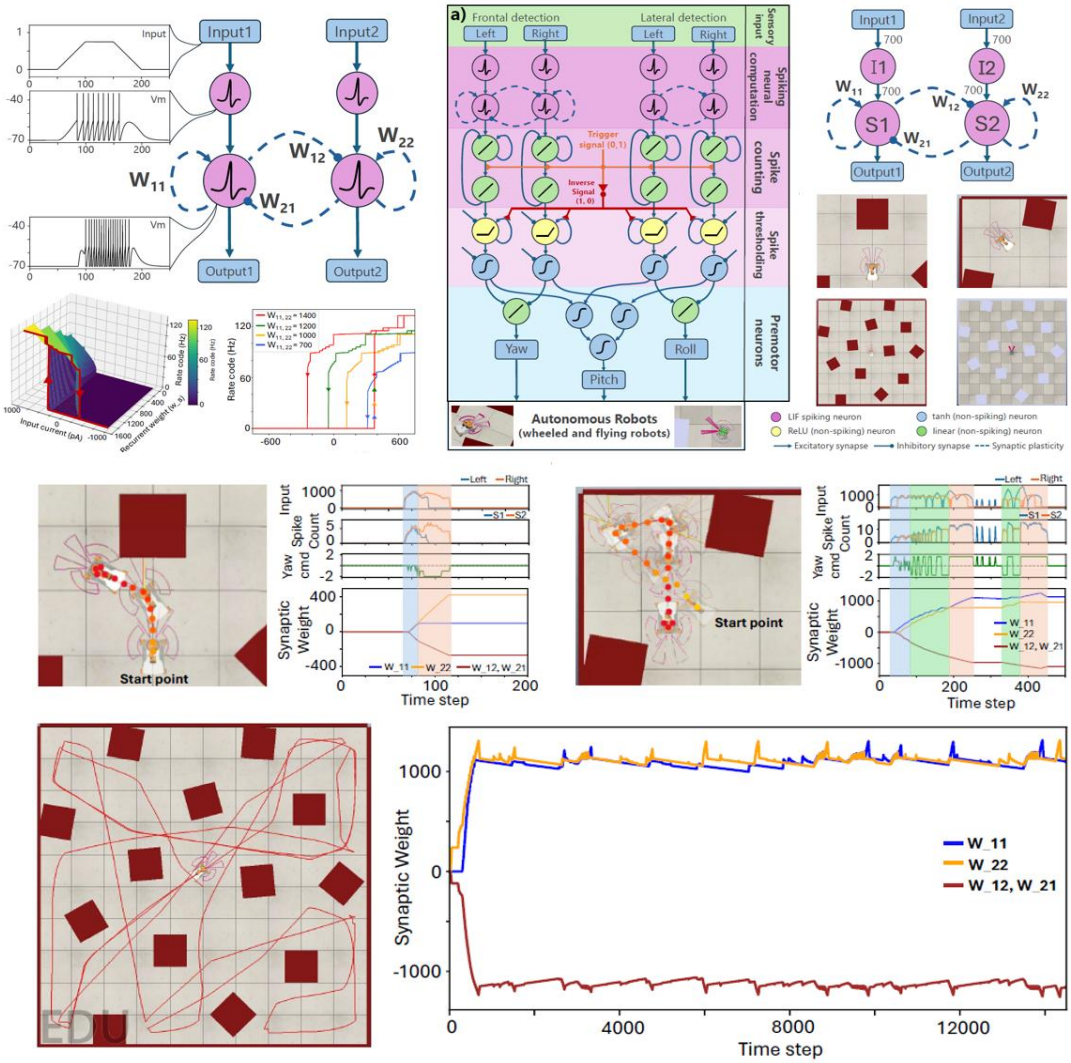

Fig. 4: *A spiking neural control system with signal dynamics at each level of the network, together with a compact neural obstacle avoidance network, and experimental results from a wheeled robot performing both short- and long-range navigation tasks in obstacle-filled environments.*

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exhibited hysteresis and short-term memory dynamics similar to those of a recurrent non-spiking neuron, while also presenting distinct characteristics, such as differences in activation thresholds and response magnitudes. Leveraging these spiking neural dynamics and online learning, I developed a compact neural control system that effectively navigated simulated autonomous robots through obstacle-filled environments.

Apart from research publications, I also applied the knowledge, skills, and neural control algorithms developed during my Ph.D. research to an industrial autonomous drone navigation control system [3]. This system integrates a multimodal object detection framework, including depth vision sensors and long-range LiDAR sensors, to detect obstacles of varying sizes at both short and long distances under different lighting conditions, including daytime and nighttime. In addition, an adaptive neural control system with online learning combines speed adaptation and adaptive obstacle avoidance (see Fig. 5). The system is implemented on an NVIDIA Jetson Orin NX onboard computer and deployed on a Tarot 650 drone equipped with a PX4 flight controller, and is integrated with the QGroundControl interface for mission planning and monitoring, resulting in a safe and efficient autonomous drone navigation system.

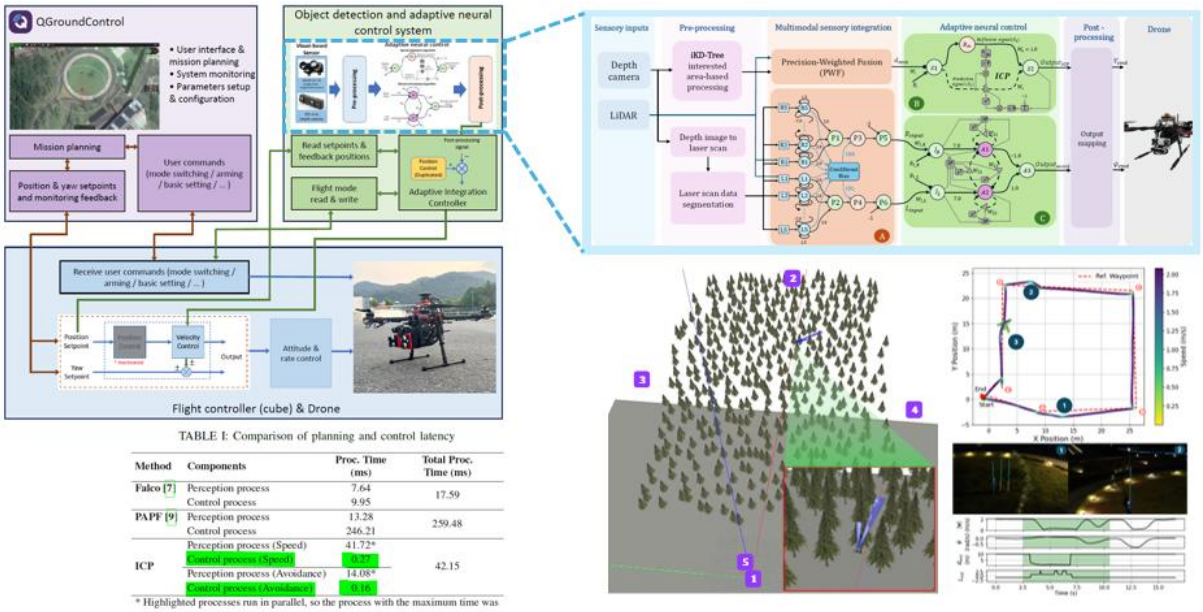

TABLE I: Comparison of planning and control latency

Method	Components	Proc. Time (ms)	Total Proc. Time (ms)
Falco [7]	Perception process	7.64	17.59
	Control process	9.95	
PAPF [9]	Perception process	13.28	259.48
	Control process	246.21	
ICP	Perception process (Speed)	41.72*	42.15
	Control process (Speed)	0.27	
	Perception process (Avoidance)	14.08*	
	Control process (Avoidance)	0.16	

* Highlighted processes run in parallel, so the process with the maximum time was selected as the total processing time.
Note that for the ICO method, the result is identical to the result of ICP method.

Fig. 5: *Overall system of the industrial autonomous drone navigation control, including the multimodal object detection module, pre-processing system, adaptive neural control, post-processing system, and the drone platform. Both simulated and real-world drone experiments are also presented.*

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During my doctoral studies, I mentored a master's student in conducting research on multi-segmented and multi-legged robots inspired by millipede locomotion. The work focused on developing a neural control system capable of propagating locomotion waves from the posterior to the anterior body segments, enabling coordinated leg movements across the robot [8]. The neural control system employed neural hysteresis dynamics to generate temporal delays among consecutive body segments from posterior to anterior, resulting in a traveling wave pattern that produces smooth and coordinated locomotion along the robot's body. Additionally, the body–leg coordination strategy investigated the synchronization between body segment activation and leg motion patterns, enabling stable support phases and efficient propulsion during multi-legged locomotion [9]. The results demonstrate the effectiveness of hysteresis-based neural dynamics in coordinating distributed body segments for robust multi-legged robotic locomotion.

The aforementioned works demonstrate my solid understanding of the fundamental principles of neural computation and learning systems, as well as my ability to effectively adapt and integrate these approaches into robotic control systems. Beyond my technical expertise, I have substantial experience in formulating original research questions, designing rigorous experiments, and critically analyzing results. Building on this background, the key research opportunities and future directions for advancing bio-inspired neural control frameworks for intelligent and autonomous robotic systems are outlined as follows.

1. Neural Control System for Autonomous Navigation and Target-Oriented Behaviors

The existing works [2, 3] can be extended to support target search and goal-directed approach by integrating additional sensors for target detection. The detected signals are processed through dedicated sensory pathways and connected to the neural control system (see Fig. 2, pink-colored part) using reverse inhibitory connections, enabling directional activation toward the detected target. This extension is currently part of ongoing work, and preliminary simulation results demonstrate that the drone can autonomously explore, detect a green target object using a color-detection sensor, and continuously navigate toward it until reaching the target.

2. Bio-Inspired Multi-Agent and Collective Intelligence Systems

The existing work [1, 2] developed a neural-based navigation and exploration system inspired by biological behavioral studies, primarily focusing on the behaviors of individual organisms. From a collective perspective, many biological systems also exhibit collective behaviors, such as fish schooling and bird flocking, which emerge from local interactions among individuals. Therefore, the existing framework can potentially be extended by integrating interaction mechanisms between agents into the neural control system of each agent, enabling collective robotic behaviors.

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3. High-Level Neural Cognitive and Multimodal Behavior Control

Beyond navigation and exploration, intelligent autonomous robots require additional capabilities such as locomotion, object manipulation, and higher-level cognitive functions including recognition, memory, planning, reasoning, and decision-making. A modular neural control architecture enables the independent development and integration of these behaviors, while a high-level behavior coordination system ensures synchronized execution across modules. This framework provides opportunities to scale toward modular cognitive coordination and multimodal autonomous operation in intelligent robotic systems.

4. Embodied and Mechanical Intelligence in Bio-Inspired Robots

Apart from efficient modular bio-inspired neural control systems, certain tasks (e.g., landing on complex terrains, mobile manipulation, or locomotion) can benefit from integrating mechanical intelligence and embodied design principles into robotic systems. Inspired by biological organisms, compliant structures and adaptive mechanical components can passively contribute to stability, locomotion, and environmental interaction. Combining such mechanical intelligence with neural control systems can significantly improve the efficiency, robustness, and adaptability of robotic platforms.

5. Biologically Plausible and Real-Time Embedded Neural Intelligence

The bio-inspired modular neural control system provides flexible functionality by allowing individual modules to be independently developed, optimized, and integrated without altering the overall architecture, enabling efficient real-time control with very low computational latency. Experimental deployments on both drones and embedded mobile robots demonstrated that the modular neural control system maintains adaptive autonomy and online learning capabilities while operating on resource-constrained hardware (e.g., an ESP32 microcontroller). Building on biological principles such as intrinsic neural dynamics, local interactions, adaptive behaviors, and online learning observed in natural organisms, this approach offers a promising foundation for scalable embedded autonomy and decentralized intelligent behavior in future robotic systems.

As a postdoctoral researcher, I aim to extend this work toward bio-inspired neural computation and real-time processing, control and learning systems for autonomous robots operating in complex and uncertain environments, as well as integrating mechanical intelligence into robotic platforms to enhance robustness, stability, and overall system performance. I am confident that my background will allow me to contribute meaningfully to your research team and advance the development of intelligent robotic and autonomous systems.

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